

Turning Up the Heat: Warmed Composites - Help or Hype?

Group 1

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MATERIAL SCIENCE ASPECTS STUDIED



The following is a list of various materials-science properties that were investigated in this report

Rheology

Viscosity

Viscoelascity

Unplugging force / work

Complex viscosity

Stickiness

Heat transfer / thermal diffusion

Monomer Conversion

Polymerization shrinkage

Monomer elution

Surface wettability

Van der Wall's forces

Behavioral Objective 1

State the relationship between composite temperature and viscosity and between resin composite temperature and its extent of monomer conversion when polymerized *at that elevated temperature*

H.K. Graham, Vaani Balyan

Search Strategy

- P: Conventional packable composites not fully adapting to restoration preparations.
- I: Pre-heating resin composite to assess a change in viscosity and polymerization
- C: Comparing to conventional room temperature resin composite
- O: The pre-heated composite will show a decrease in viscosity and increase in monomer conversion that improves restoration success.

Databases used

Pubmed

Keywords

Resin composite, temperature, viscosity, monomer conversion

Pubmed search string

((composite)) AND ((temperature)) AND (viscosity)

((composite)) AND ((temperature) AND ((polymerization))

Search results

256 results

5 used

Highest level of evidence

Randomized clinical trials





Pre-heating leads to a drop of 30-82% in the viscosity of packable materials (Loumprinis et al., 2020)

Study comparing packable composite viscosity change vs flowable composite when pre-heated

Also showed a reduction in stickiness to dental instruments at higher temperatures

Both time and temperature are statistically significant predictors of viscosity (Lucey et al., 2010)

At a temperature of 24 °C, the model of viscosity against time was $\text{viscosity} = 660.2 + 1.09 s$. At a temperature of 60 °C, the model of viscosity against time was $\text{viscosity} = 261.4 + 1.09 s$.

Pre-warming composite to 68°C before insertion led to multiple structural improvements (Bolding et al., 2023)

Better adaptation, fewer large porosities, and a modest improvement in diametral tensile strength



Reviewing the science

At elevated temperatures, thermal vibrations force monomers of the matrix further apart and allow them to slide past one another more easily, which lowers the internal friction





A strong positive correlation existed between temperature and monomer conversion (Doronch et al., 2005)

Relative to the control, top-surface conversion increased 36% at 60° C, and at the 2 mm level increased 38% (also at 60° C)

Increasing composite temperature allows for lower polymerization stress while maintaining or increasing the degree of conversion (Calheiros et al., 2014)

Only when the composite temperature was raised to 60°C was a significant decrease in stress noted, giving evidence to the potential impact of using pre-heated composites



Reviewing the science

The kinetic parameters for polymerization of resin monomers have been shown to follow an Arrhenius-type behavior such that a relatively modest increase in temperature may promote a large increase in reaction rate

Aside from increased temp lowering viscosity, it enhances radical mobility which results in higher levels of polymerization and less leftover unreacted monomers

Increased temperature increases free volume, which gives radicals increased mobility to move and react with each other

IN SUMMARY:

 Resin Composite Temperature =  Viscosity

 Resin Composite Temperature =  Monomer Conversion

Behavioral Objective 2

State if heating conventional resin composite reduce its viscosity to the level of flowable composite?

Loren Wray

Search Strategy



P: Conventional dental resin composite

I: Heating/preheating composite

C: Flowable composite

O: Reduction in viscosity

Databases used

Pubmed

ChatGPT

Keywords

Conventional dental resin composite, Flowable composite resin, Preheated composite, Warmed composite, viscosity, rheological properties

Pubmed search string

("resin composite" OR "dental composite" OR "conventional composite")

AND ("heating" OR "preheating" OR "warming")

AND ("flowable composite" OR "low viscosity composite")

AND ("viscosity" OR "flowability" OR "rheological properties")

Background Info



What are differences between resin composite and flowable composite?

CONVENTIONAL RESIN COMPOSITE: Higher filler content = stronger but more viscous

↓ Resin component and higher filler loading = ↓ polymerization shrinkage

FLOWABLE COMPOSITE: Lower filler content = flows easier but mechanically weaker

↑ Resin component and lower filler loading = ↑ polymerization shrinkage

What does heating do? Can reduce viscosity and improve adaptation

WHAT IS VISCOLOR THERMO-VISCOUS COMPOSITE?

Acts as a flowable composite when warmed and added to the restoration, then quickly cools to immediately be sculpted as a conventional, packable composite

RELATED LITERATURE



Rueggeberg et al., found that the film-forming thickness was reduced by 30% when the resin was heated to 54 °C, showing reduced viscosity and improved fluidity composite resins at high temperatures.... But flow resin applications cannot be replaced by preheated conventional composite resins. (Bhopatkar, 2022)

Blalock reported that preheating conventional composite resin yields lower film thickness for some products, but flow cannot be attributed to composite resin classification, filler content, or shape. Preheated composite resin thick-ness was greater than that of all flowables. (Blalock, 2006)

Baroudi found that preheating composite reduces viscosity and increases flowability, which facilitates better adaptation to cavity walls. This reduces microleakage and results in superior marginal adaptation (Baroudi, 2015).

Material Type	Temp (°C)	Viscosity (kPa·s)	Observation
Packable composite	23	6.75–19.14	Thick, poor flow
Flowable composite	23	1.31–2.20	High flow
Heated packable composite	45-54	2.025-15.7 (↓ 30–82%)	Improved handling, not as fluid as flowable
Thermo viscous formulations	45-54	≈ Flowable	Comparable flow only in special formulations

Summary of B02



Heating will significantly reduce viscosity, narrowing the viscosity gap, but it does not eliminate it

Most conventional composite resins remain **more viscous than flowable composites** due to the difference in filler content

Only exception is the VisCalor thermo-viscous formulation, which is comparable to flowable composite when warmed.



Behavioral Objective 3

State the factors influencing the viscosity and adaptation of preheated composite to the internal preparation walls when it is delivered into clinical preparations.

Daniel Ruiz, James Valdez

Search Strategy



P: Pre-heated composite vs Room temperature composite in effectiveness in adaptation and how that influences viscosity of material

I: Pre-heating composite when delivered to cooler temperatures

C: Comparing factors influencing viscosity and effectiveness in adaptation to tooth surface

O: Several factors certainly influences viscosity/ adaptation of the internal walls.

Databases used

PubMed

MDPI

Highest level of evidence

Randomized control trials

QUERY via Microsoft AI Copilot search string:

“Please give me literature references from highly indexed peer-reviewed journals that have online pdf articles either freely available or by subscription related to the topic of the **factors influencing the viscosity and adaptation of pre-heated composite when placed on a prepared tooth in clinical situations.** Within each reference, please provide further details on **cooling kinetics and viscosity changes of warmed composite.** Please list each article found with annotation references using Vancouver styles.”





The effects of stickiness as viscosity changes with increasing temperature

STICKINESS: ability of a resin composite to adhere to the surface of a dental instrument for handling and increased marginal adaptation.

At a molecular level, adhesion between placement instruments and composite occurs via Van der Waals forces (weak)

Inverse correlation between wettability and viscosity. (e.g. Reduced resin composite wettability at higher viscosity means reduced resistance to detachment) (Loumprinis et al. 2020)

STICKINESS measured via vertical detachment based on the values of unplugging force over unplugging work.

UNPLUGGING WORK: the total energy required to unplug the instrument (remove instrument from material).

UNPLUGGING FORCE: the force required to initiate flow within a material between two surfaces. In other words, it's the minimum stress needed to “unplug” or start movement of a material.

COMPLEX VISCOSITY: A form of **VISCOELASTICITY**. Measures a materials' resistance to deformation under oscillatory shear, includes both the elastic (energy storage) and viscous (energy dissipation) components of the fluid's response (Loumprinis et al. 2020).

OVERCOMING FLOW RESISTANCE OF COMPOSITE IN COMPULE

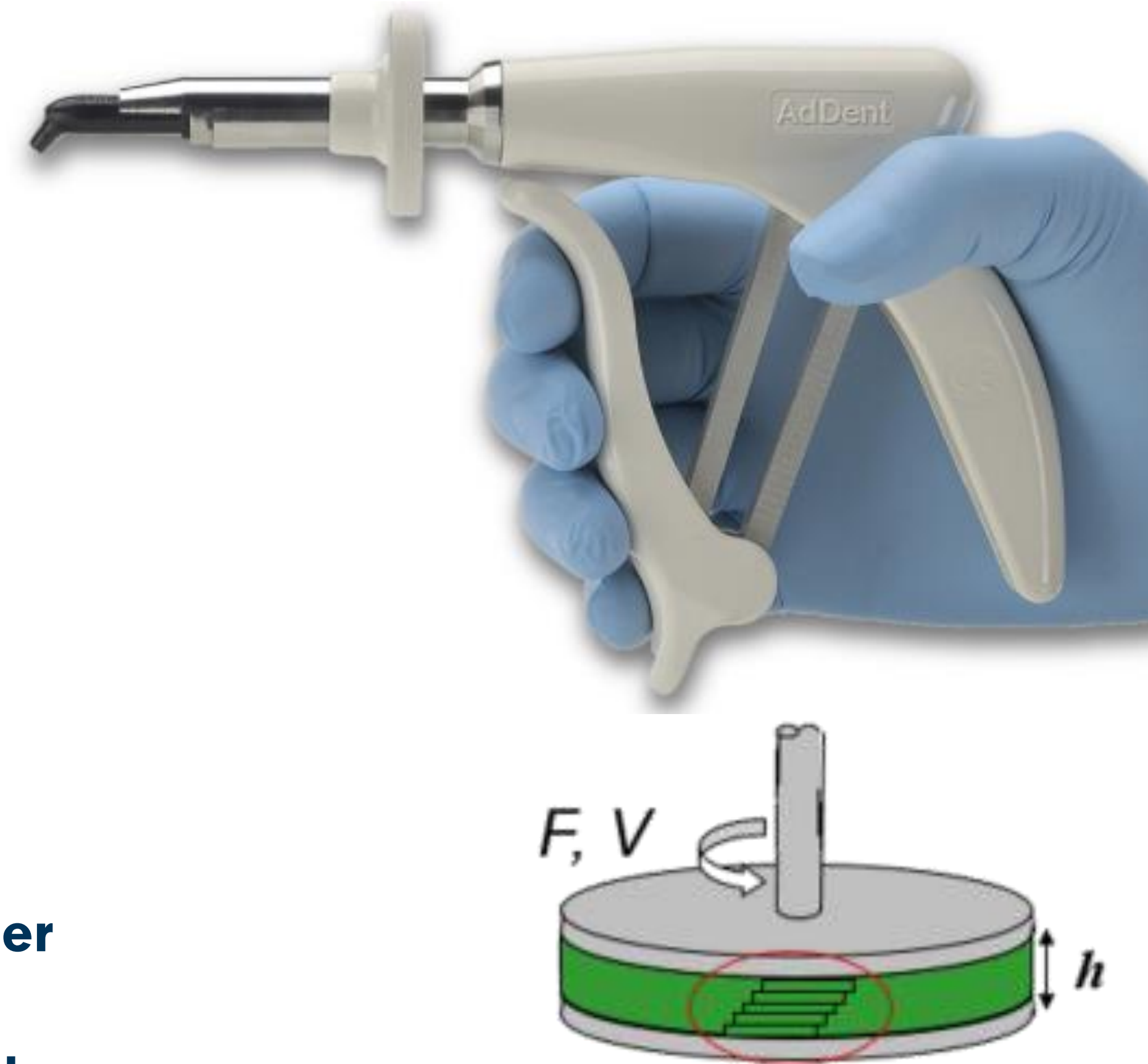


Figure 2 - Illustration showing a sample loaded between parallel plates and shear profile generated across the gap
2016 Malvern Instruments Limited

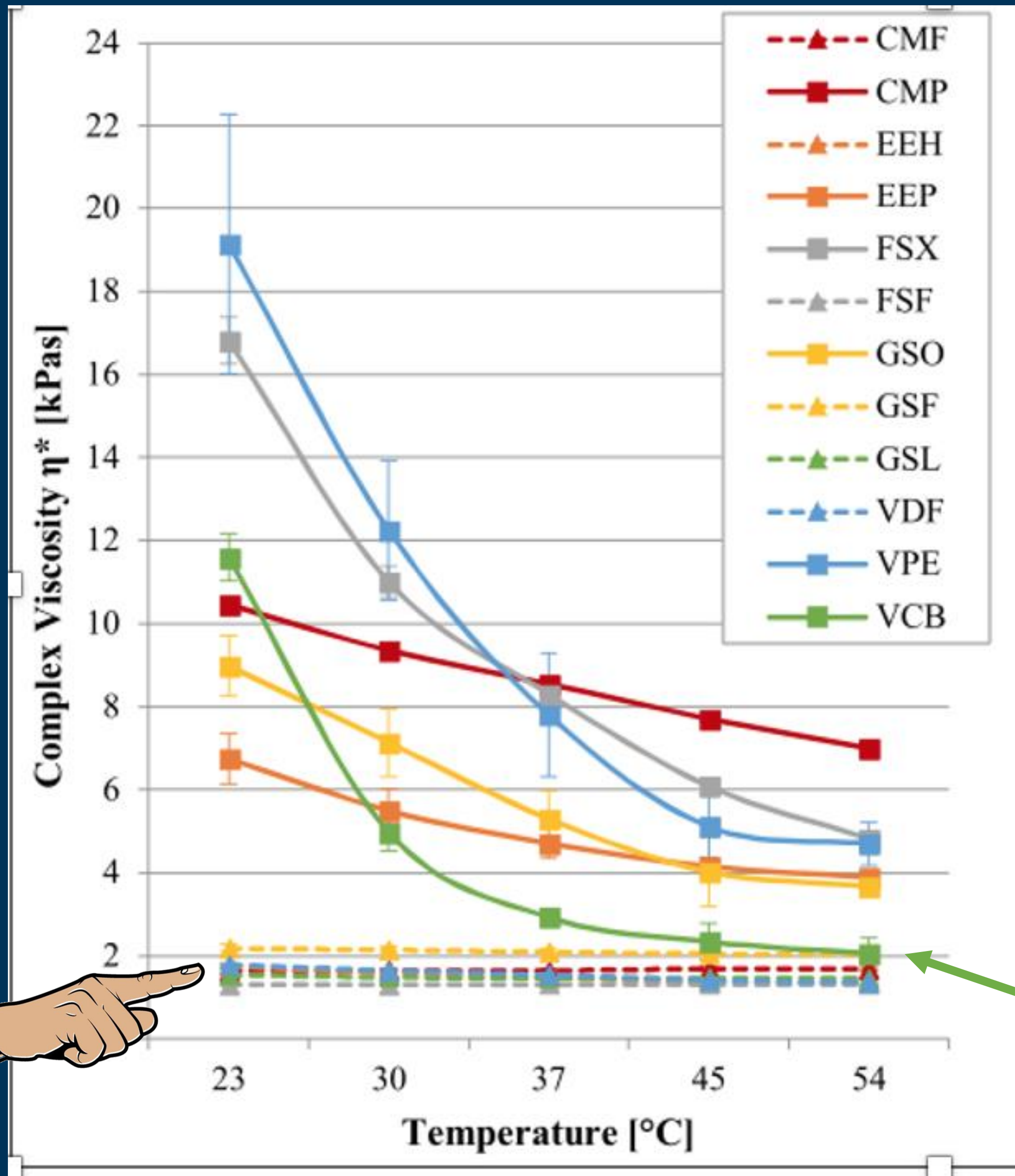
Comparison of flowable vs packable composite viscosities at increasing temperatures

Dotted lines = flowable composite types – **do NOT change with temperature**

“Increasing the temperature from 23 to 54°C, condensable resin composites are reaching the viscosity of their flowable counterparts but offering improved adaptation in an oral cavity.” (Loumprinis et al. 2020)

Temperature effect is highly material-dependent

VisCalor Bulk: BisGMA/aliphatic dimethacrylate-based thermo-viscous RBC



Force-rotational distance curves for room temperature and pre-heated packable composite (stickiness)

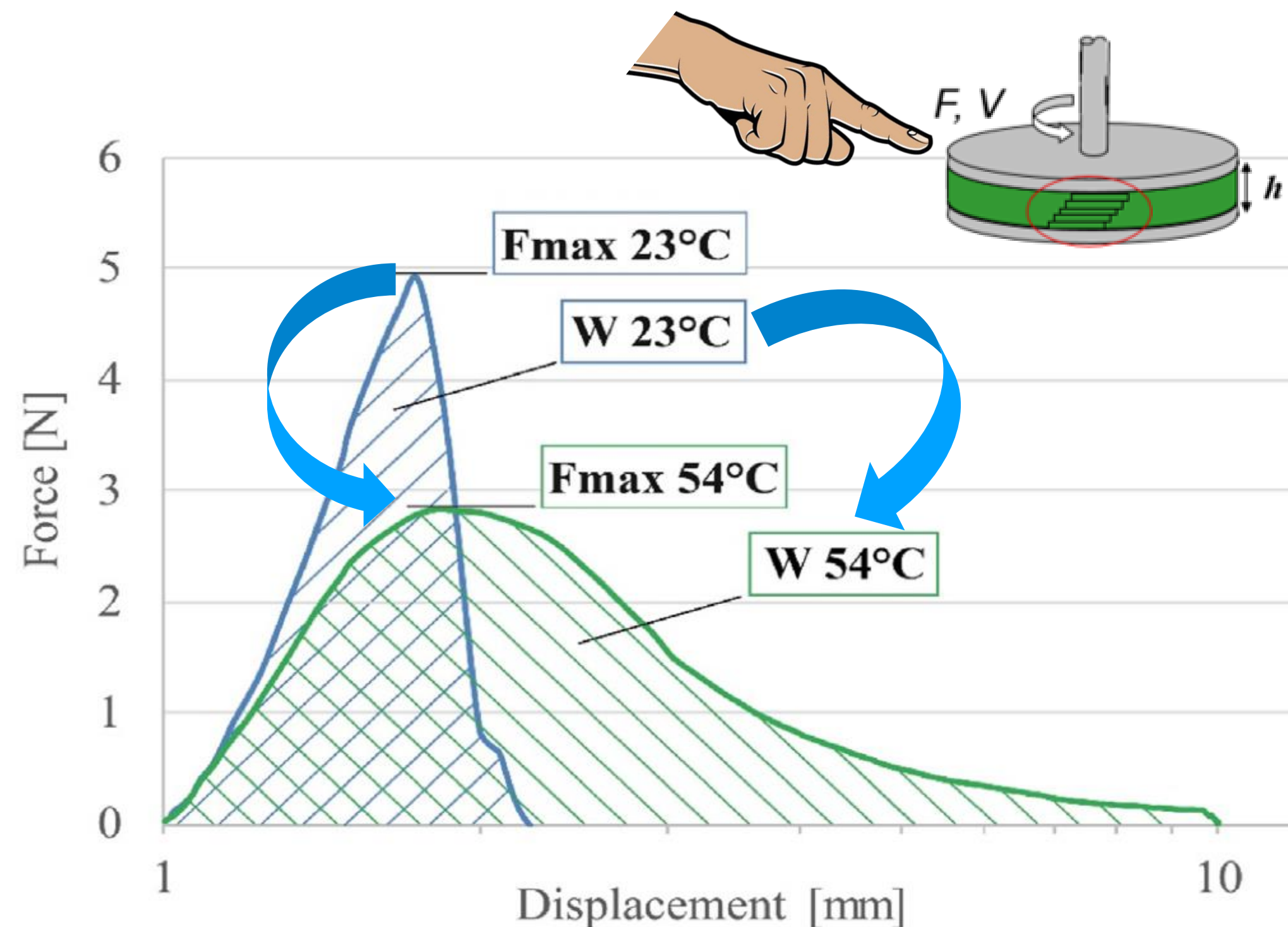


Fig. 5 – Schematic depiction of force-displacement diagrams of the same material at two different temperatures (23/54°C) showing the differences between maximum unplugging force (peak) and unplugging work (area).

Six Different Packable Composite types heated on a 54°C in a plate-plate rheometer to determine viscosity. Normal force and unplugging work (stickiness) noted.

Larger unplugging work AREA under the curve at increasing temperatures (i.e. lower viscosity).

At 23°C, maximum unplugging FORCE (Fmax) is higher than at 54°C, but the area under the curve (unplugging work) is larger for the measurement at 54°C.

At higher temps, the upper plate is in contact with the material for a longer time, so the composite in turn stretches more and is stickier, regardless of the maximum force needed to separate (Loumprinis N, et al. 2020).

How does monomer elution (release) change upon pre-heating of different composites?



DEGREE OF MONOMER CONVERSION: The percent of monomer double bonds ($C=C$) that are converted into single bonds ($C-C$) during the polymerization (curing) process of a resin-based material

MONOMER ELUTION: is the leaching of residual unreacted monomers from polymerized resin materials, influenced by how well the resin was cured and affecting both biocompatibility and durability

IMPOSSIBLE to have 100% degree of monomer conversion because residual monomers from polymer can be leaked due to fluids/solvents

Monomer elution compared between room temperature (23) and pre-heated temperature composites using reverse high liquid chromatography (RP-HPLC).



BisGMA, UDMA, TEGDMA were monomers and elution rates were recorded (Dunavari et al. 2022)

Products compared:

Filtek 1 (conventional bulk-fill)

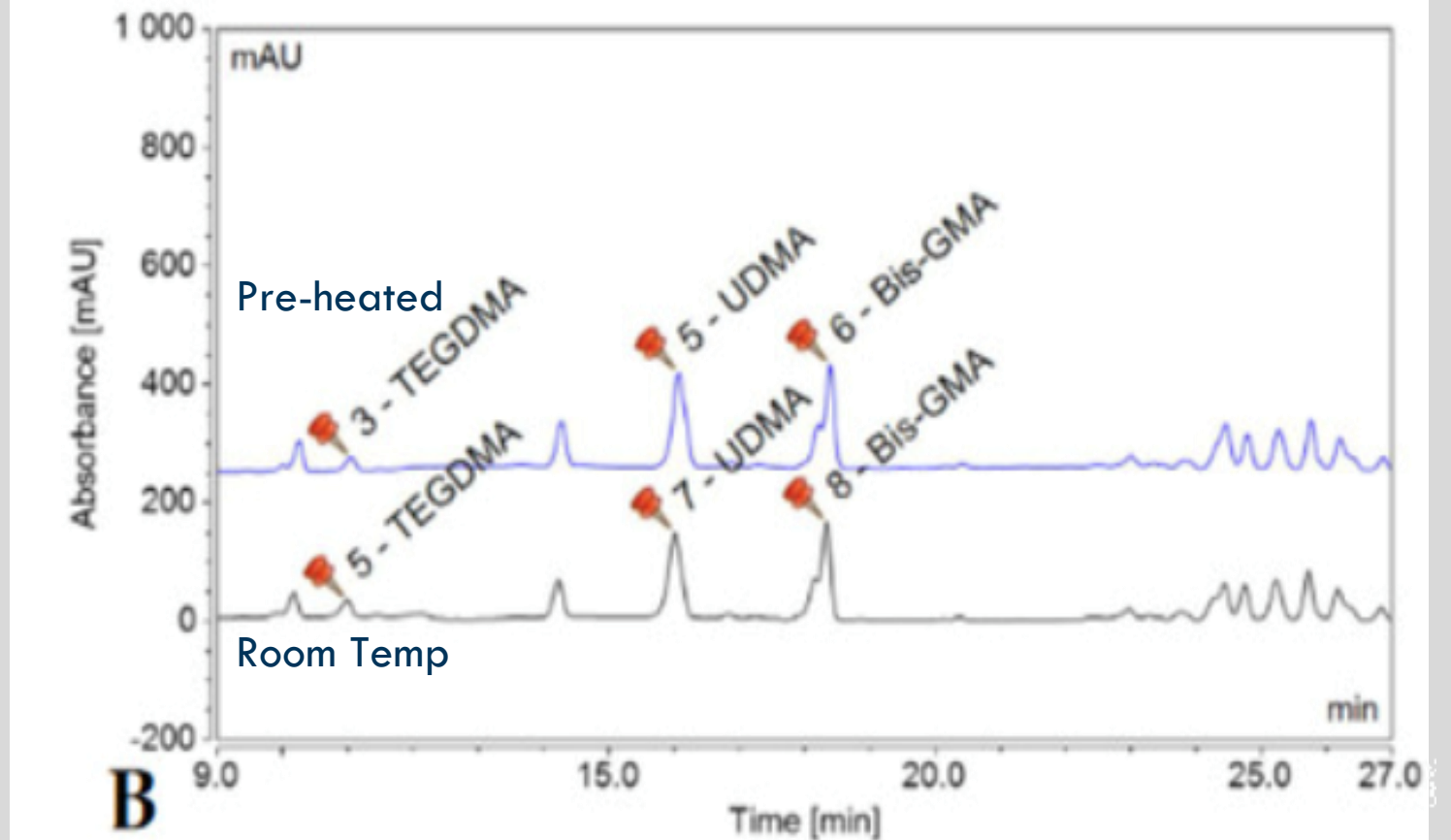
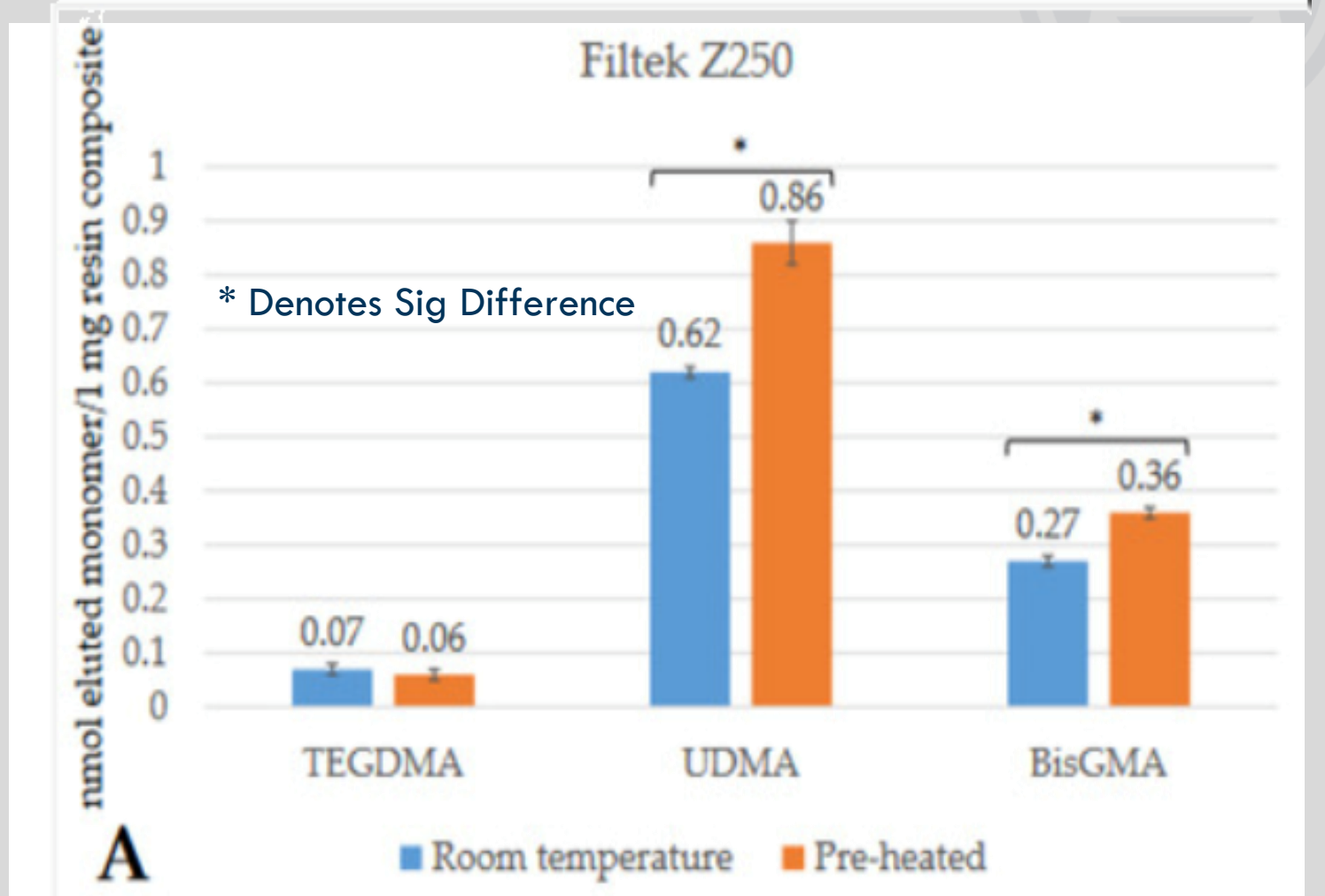
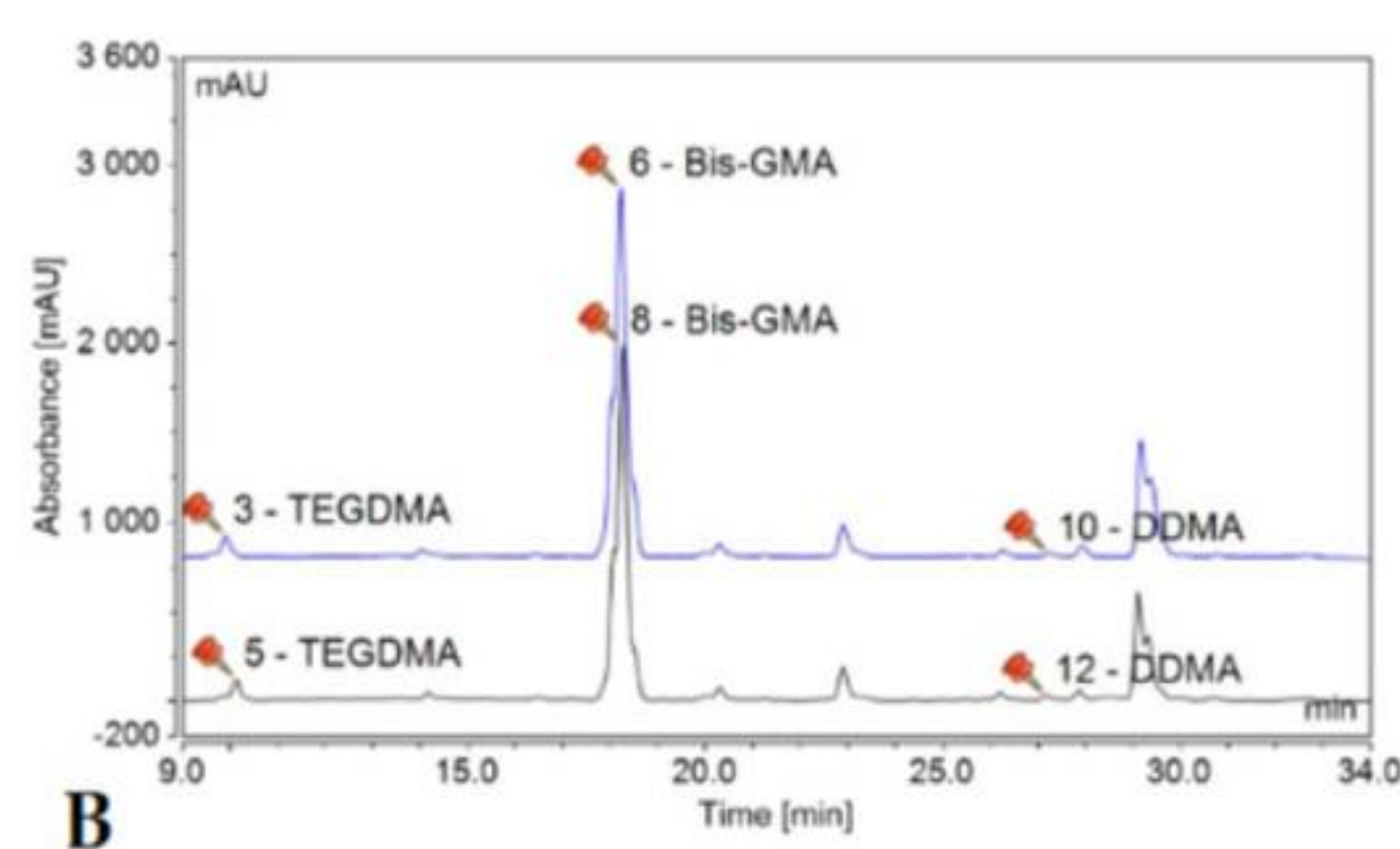
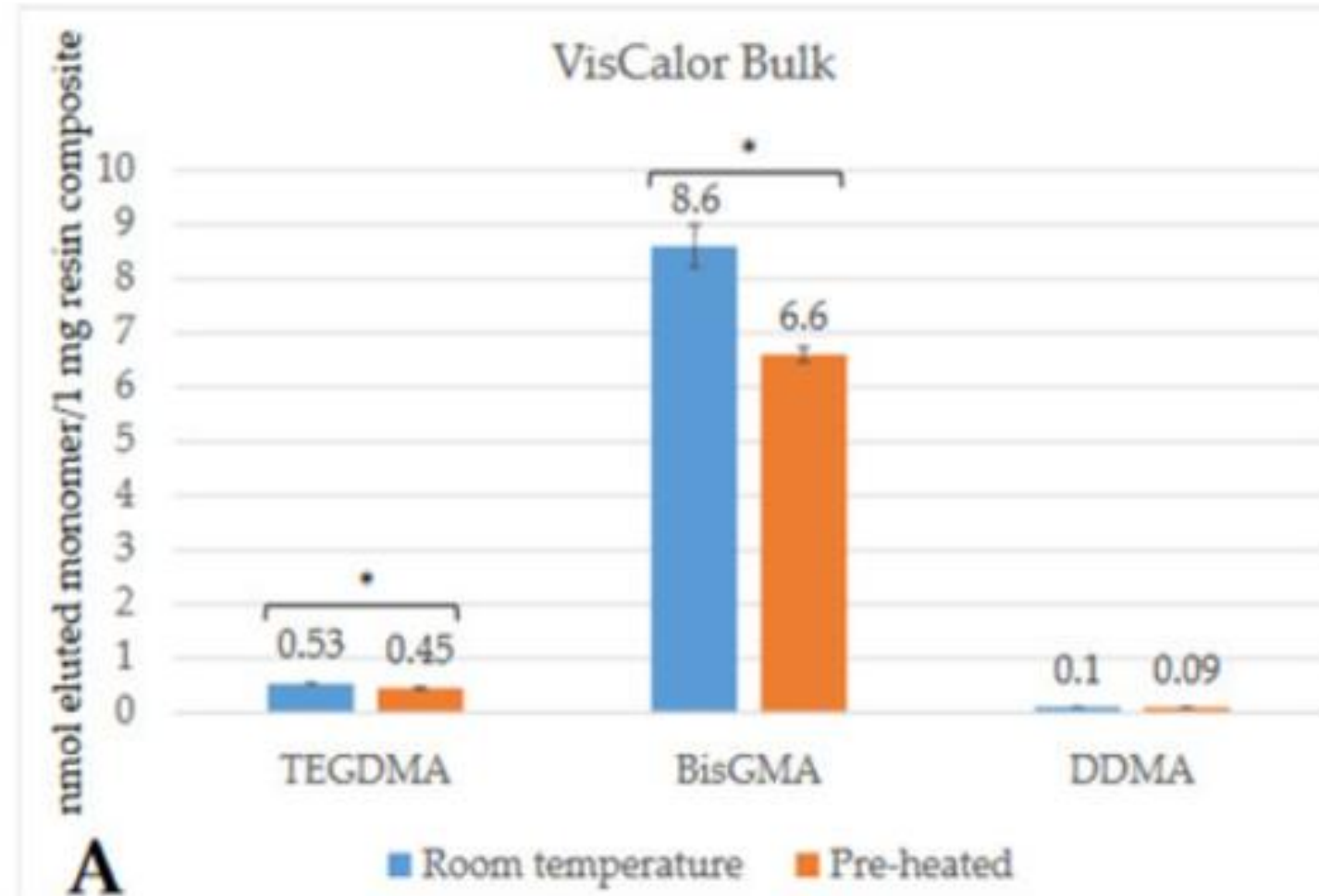
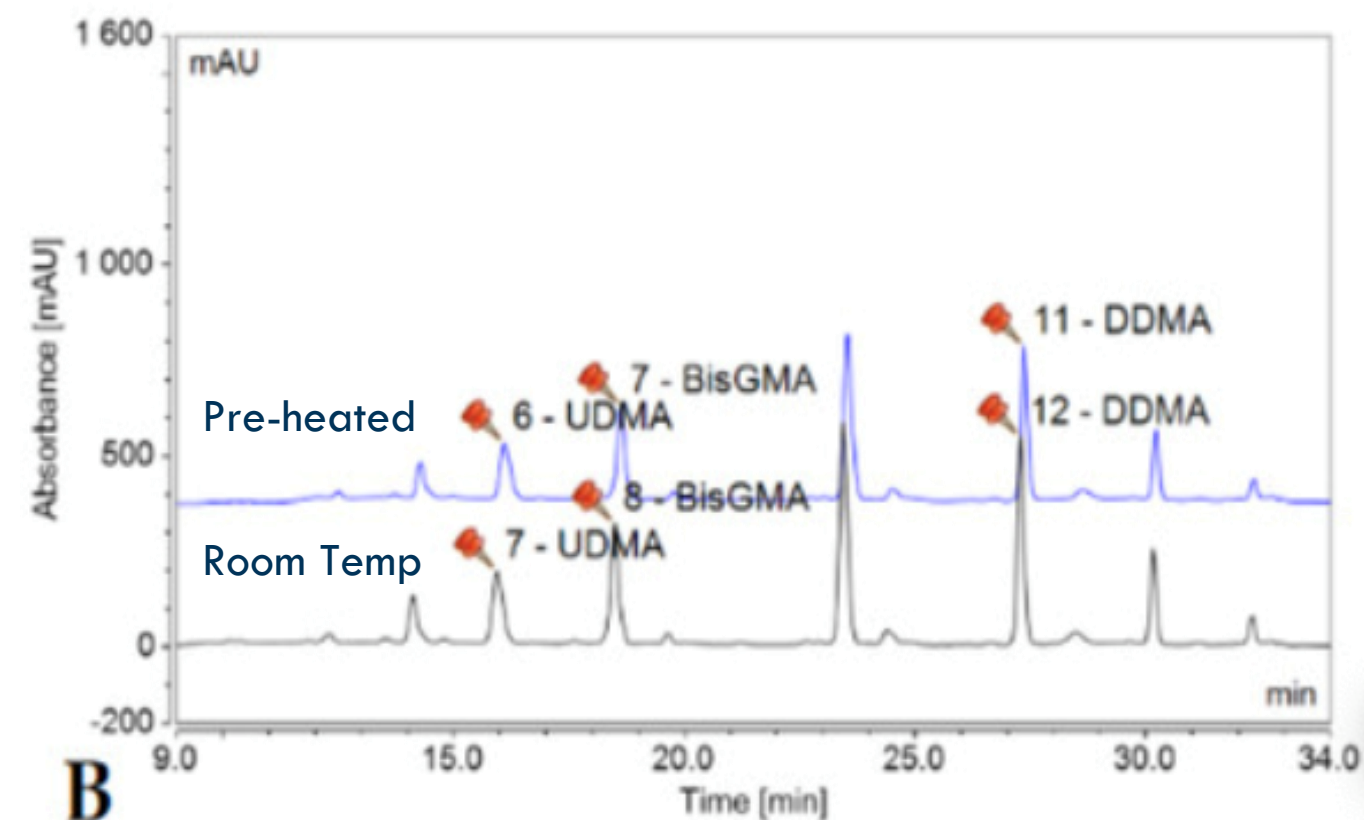
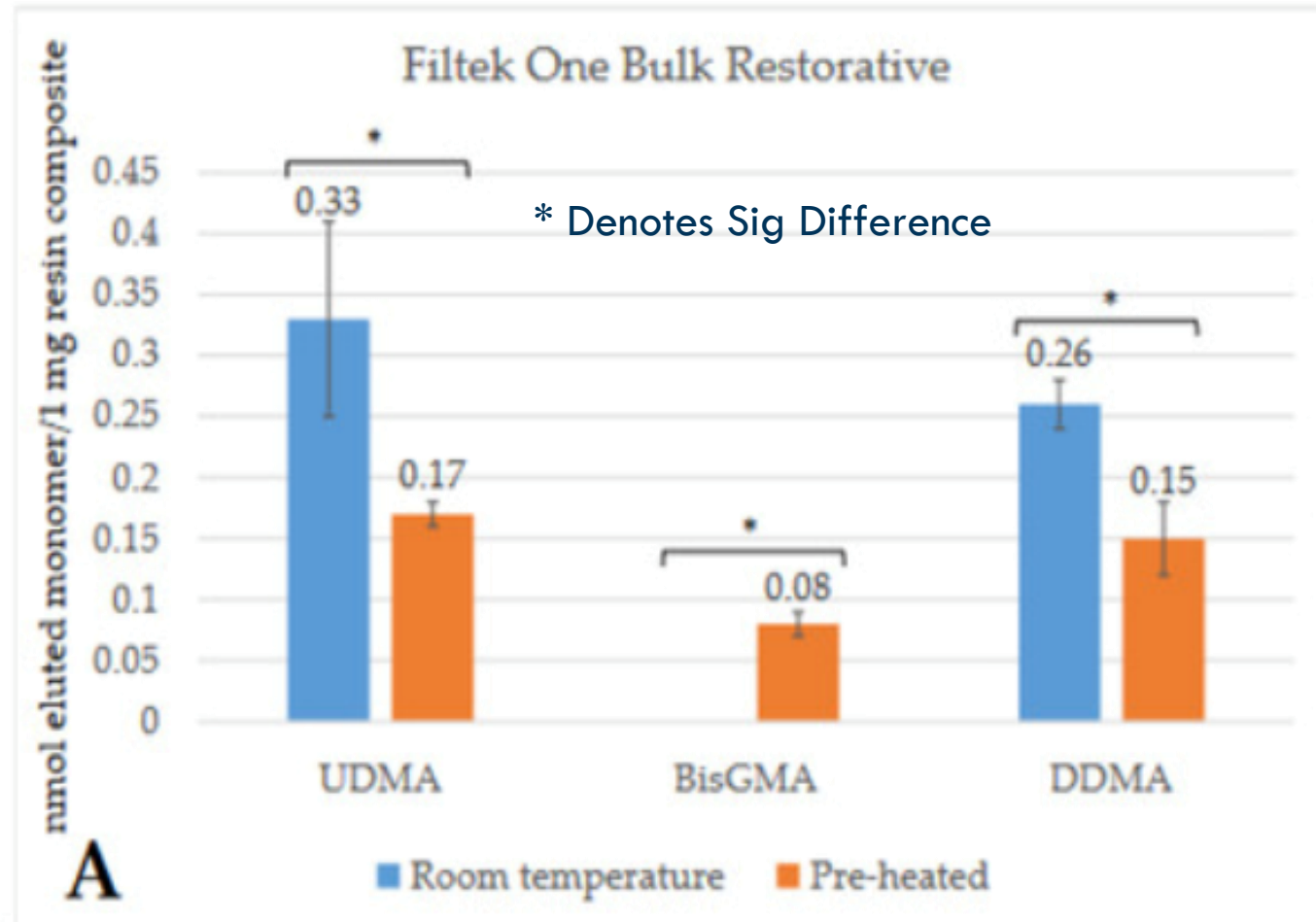
VisCalor (bulk fill designed specifically for pre-heating)

Filtek Z250 (conventional layered composite)

Monomer Elution Comparison at Increasing Temperatures

Filtek, VisCalor Bulk and Filtek Z250

(Dunavari et al. 2022)





TAKE-AWAYS

Units of monomer elution values change significantly with increasing temperature

Preheating → ↓ viscosity, better handling.

Bulk-fill composites: preheating = ↓ monomer elution (good for biocompatibility).

Conventional composites: preheating = ↑ monomer elution (less favorable) & porosity.



How does viscosity affect the marginal Integrity/film thickness of dental composite at increasing temperatures?

Marginal gaps can be caused by monomer → polymer conversion between tooth and adhesive material (polymerization shrinkage)

Flowable composites have less filler content and is a good intermediate; however, questionable durability against cyclic occlusal wear on posterior restorations.

Does the application of pre-heating composite \on the tooth surface improve the marginal integrity of the restoration?





Marginal Integrity and Film Thickness of Composite

RCT: 40 extracted human molars → Class II cavities prepared (4 mm length x 2 mm axial wall x 4 mm height)

4 groups (n=10 each):

- 1. Grandio Nanohybrid composite (80% filler) (room temp 23 °C)**
- 2. Preheated Grandio Nanohybrid composite (80% filler) (60 °C)**
- 3. Flowable liner (87% filler) + Grandio Nanohybrid composite**
- 4. Flowable liner (87% filler) + Pre-heated Grandio composite**

Evaluation: SEM analysis after thermocycling to determine Gap length (length gap/ length margin x 100%)



Findings



Group 1 (room temp composite): Highest gap (~ 27%)

Group 3 (flowable + composite): ~ 24%

Group 2 (preheated composite): ~ 16%

Group 4 (flowable + preheated): Lowest gap ~ 13%

“Preheating composites is an effective way to improve adaptation, reduce microleakage, and increase restoration longevity. Flowable liners alone don’t provide this benefit, but preheated composites act like a flowable while maintaining mechanical strength.” (Elkady et. Al 2020).

Summary



Preheating composite lowers marginal gaps present with lower film thickness = lower viscosity at rising temperatures. Best adaptation was seen with preheated composite, with or without a flowable liner.

Preheating also increases monomer mobility, leading to higher monomer conversion, which improves adaptation and bond strength with quick handling upon tooth surface placement (Elkady et. Al 2020).

Study limitations - tooth specimens were extracted. The experimental conditions differed greatly in temperatures / environment compared to intraoral circumstances.

(i.e., not applied to tooth surface of in-vivo conditions)

Summary of B03



Several factors influence viscosity and adaptation to the internal walls of a prepped tooth with preheated composite:

increased wettability, decreased monomer elution in bulkfill composites, decreased film thickness, and increased stickiness to provide increased adaptation and workability.



Behavioral Objective 4

Identify methods used to pre-heat resin composite either prior to or subsequent to its placement into a clinical tooth preparation and limitations/advantages to using the various methods.

Anna Claire Parviainen Ruiyang Zhao

Search Strategy

P: Patient needing resin composite restoration

I: Preheated resin composite

C: Resin composite at room temperature

O: Change in pulpal temperature; Change in restoration quality

Databases used

Pubmed

Google Scholar

Keywords

Preheated resin composite, pulp, temperature, heat

Pubmed search string

(Preheated Resin Composite) AND (Temperature)

(Preheated Resin Composite) AND (Performance)

Filters: 10 years, randomized controlled trials, meta-analysis, systematic reviews, reviews

Search results

1st search string: 9 results

2nd search string: 12 results

6 were used for this project

Highest level of evidence

Meta-analysis



WHAT'S A SAFE TEMPERATURE FOR THE PULP?

Zach and Cohen reported that an increase of 5.5°C in temperature can result in irreversible pulpitis (intra-pulpal temperature $\geq 42.4^{\circ}\text{C}$)

- Study done on monkey teeth with soldering iron as heat source

WILL THAT TEMPERATURE BE REACHED?

Pulpal blood flow acts as a heat sink

Dentinal fluid flow enhances the heat transfer

Dependent on many factors, hard to test *in vitro*

- Water content in the pulp
- Changes in pulp blood and dentinal fluid flows
- Previous injury to the pulp
- Remaining dentin thickness and insulating quality
- Duration of insult
- Surface area of exposed dentinal tubules



In vitro Experiment (Hatner, 2024)

Using preheated resin for veneer cementation on bovine incisors at room temperature

Group with 2.0mm preparation depth with preheated composite resin had a mean intrapulpal temperature increase of 5.70 ± 2.14 °C

In vivo Experiment (Rueggeberg, 2010)

When stored in the commercial device set to 60°C, the composite contents only reached an average of 54.7°C

As the preheated composite was injected into the tooth preparation, the maximum temperature recorded was only 38.4°C (instead of 54.7°C)

Preheated composite rapidly cools down after being removed from the heating device

Does preheated resin composite offer superior restoration properties?



36-Month Randomized Clinical Trial Evaluation of Preheated and Room Temperature Resin Composite (Elkaffas, 2022)

Preheated nanofilled RCs showed an acceptable clinical performance with better resistance to marginal staining.

The clinical performance of dental resin composite repeatedly preheated: A randomized controlled clinical trial (Elkady, 2024)

One and ten times of preheating conducted better clinical performance in marginal adaptation, marginal discoloration, and color matching

A systematic review and meta-analysis on using preheated resin composites as luting agents for indirect restorations (Barbon, 2022)

**Preheated RCs showed significantly higher film thickness compared to resin cements →
DO NOT USE**



Considerations when using preheated resin composite clinically:

Compules will not heat to theoretical temperature

Composite rapidly cools down after being removed from heat source

Avoid if RDT is thin/pulp capping

What we should look for in our products:

Continuous, controlled-heating of delivery device

Compules/composites that can withstand repeated heating

Heating Method	Indications and Directions for User	Drawbacks	Compatible Resin Form Factors	Examples of Products
Nascimento Poubel et. al., 2022				
Heated Dispenser	Maintains heat during transfer and placement, ideal for direct restorations, no loss of time	Expensive	Compules	Compex HD Composite Gun, VisCalor Dispensor, Phasor™ Composite Warmer
Benchtop Composite Warmer	Place compule or gun into the warmer a few minutes before use, set at recommended temperature	Expensive, Loss of heat during loading	Compules, guns, tubes, syringes	Calset Composite Warmer, C-Flo Pro, Therma-Flo™
Digital Wax Pot	Less expensive than warmers specifically manufactured for composites	Loss of heat during loading, outside of recommended use for product	Compules	Besqual S200 Digital Wax Dipping Pot, SJK Digital Wax Pot
Overhead Light	Inexpensive, only possible with halogen lights or heat-producing lights	Difficult to control temperature of composite	Any form	
PinkWave™ LED Curing Light	If heats successfully, would be least disruptive to workflow and least time consuming	Expensive, Not manufactured to be a way to increase temperature of composite of restoration, unclear if temperature increases to beneficial amount	Any form	



Limited Clinical Data Comparing Methods

Randomized Clinical Trial Evaluating 24-Month Survival of NCCL Restoration using Two Different Methods of Thermoviscous Composite Resins (Favoreto et. al., 2024)

Compared Heated Dispenser (VisCalor) and Bench Heater (Caps Warmer)

Two restorations from each group lost, no significant difference in retention rate or other FDI criteria

Shorter working time reported for the heated dispenser



Heat composites only if recommended by manufacturer

3M Filtek Line

Flowable syringes up to 70°C for up to 1 hour, up to 25 times

Heat capsules to 70°C for up to 1 hour

Specifically says to not heat from lightbulb or unapproved heating device

VisCalor Bulk Thermoviscous Composite

Heat in Viscolor Warmer and apply within 20s, temperature should still be 61°C

Select 65°C setting if using VisCalor Dispenser

Some brands of resin composite may contain heat-cured materials

i.e. Clearfil AP-X from Kuraray- contains coumarins and benzoyl peroxide

Noticed increased in viscosity after heated more than 10-15 min (Magne, Rueggeberg)

Behavioral Objective 5

Compare the time required for a resin composite compule contents to reach CalSet-equilvanet temperatures using a halogen operator light determine the halogen light-heated composite's maximum temperature, and state how long it takes for operator light-heated composite to cool to a value similar to that of composite pre-heated using the CalSet device.

Anna Claire Parviainen, H.K. Graham, Ruiyang Zhao

How long does it take to heat up resin composite using a bench top heater vs overhead halogen lamp?

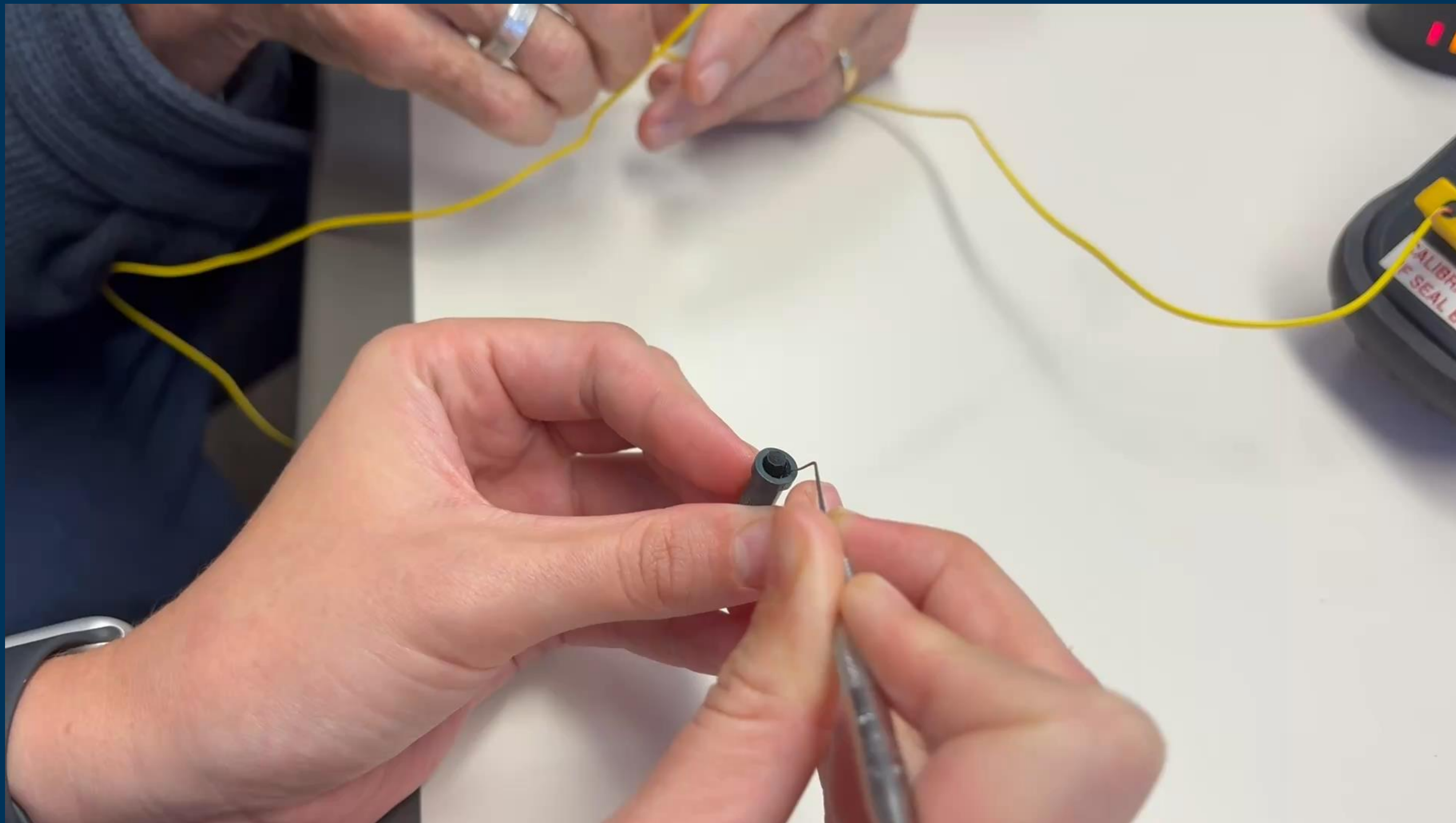


CalSet Heater (AdDent)



Overhead Halogen
lamp



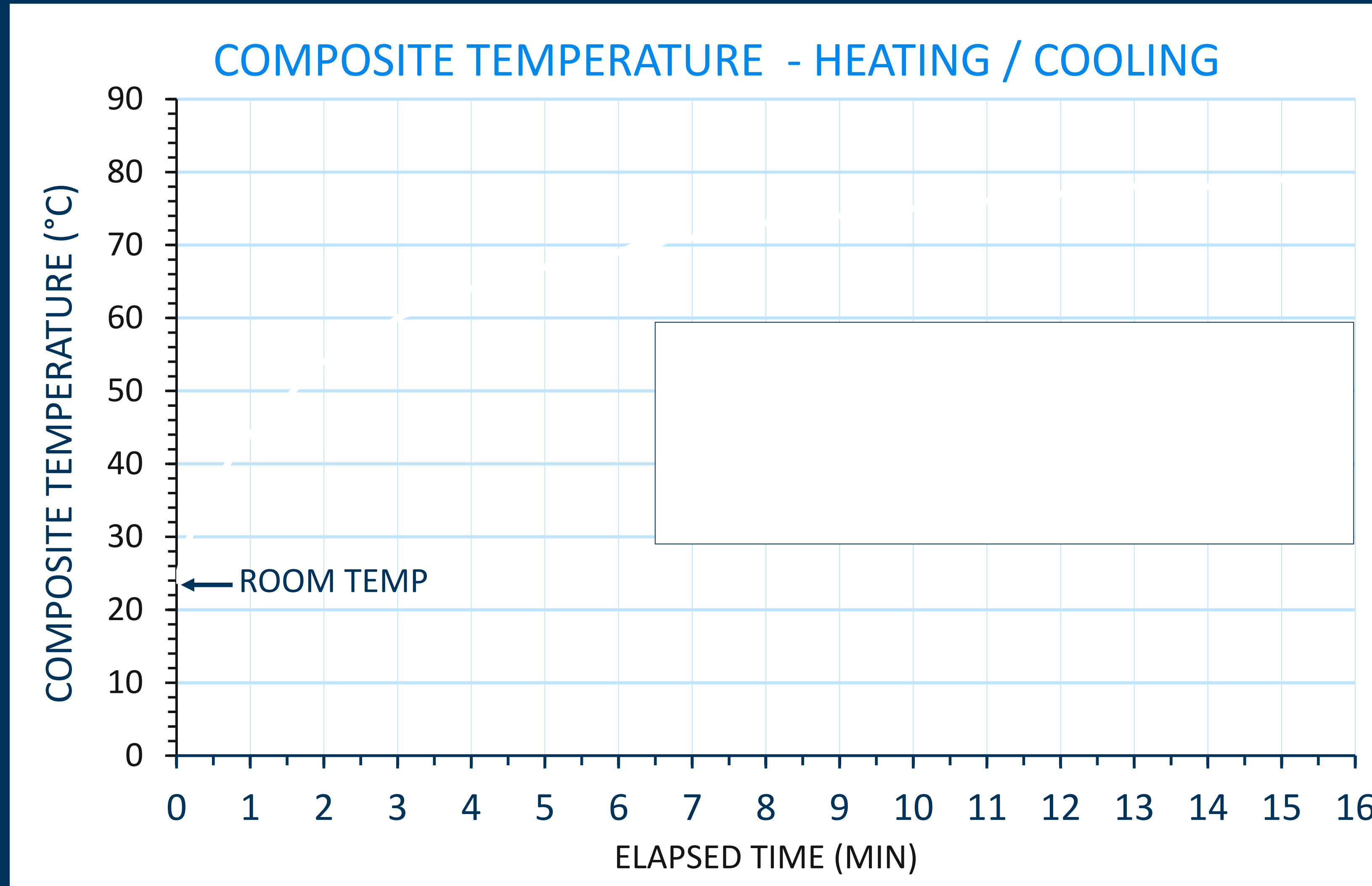


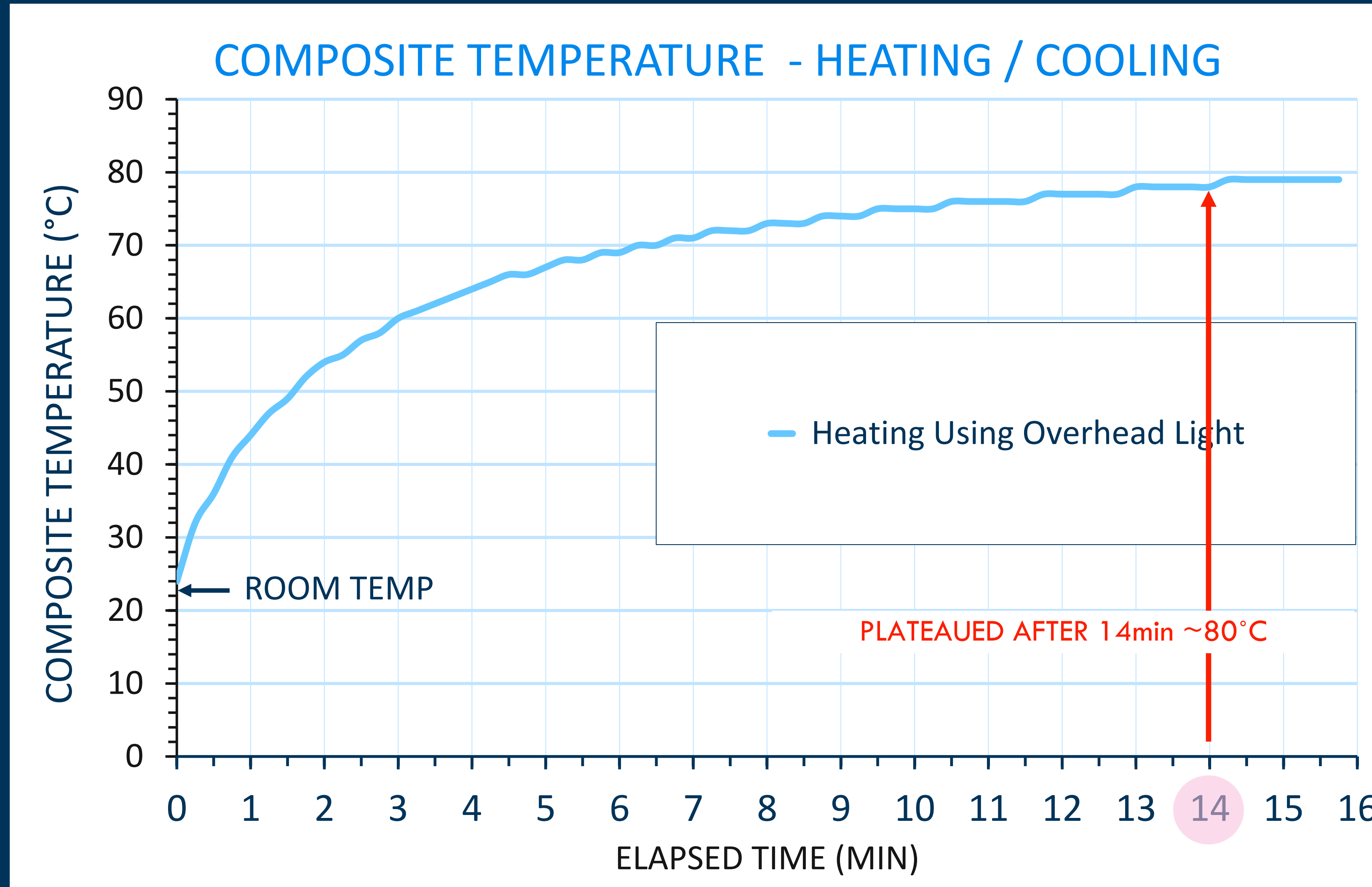


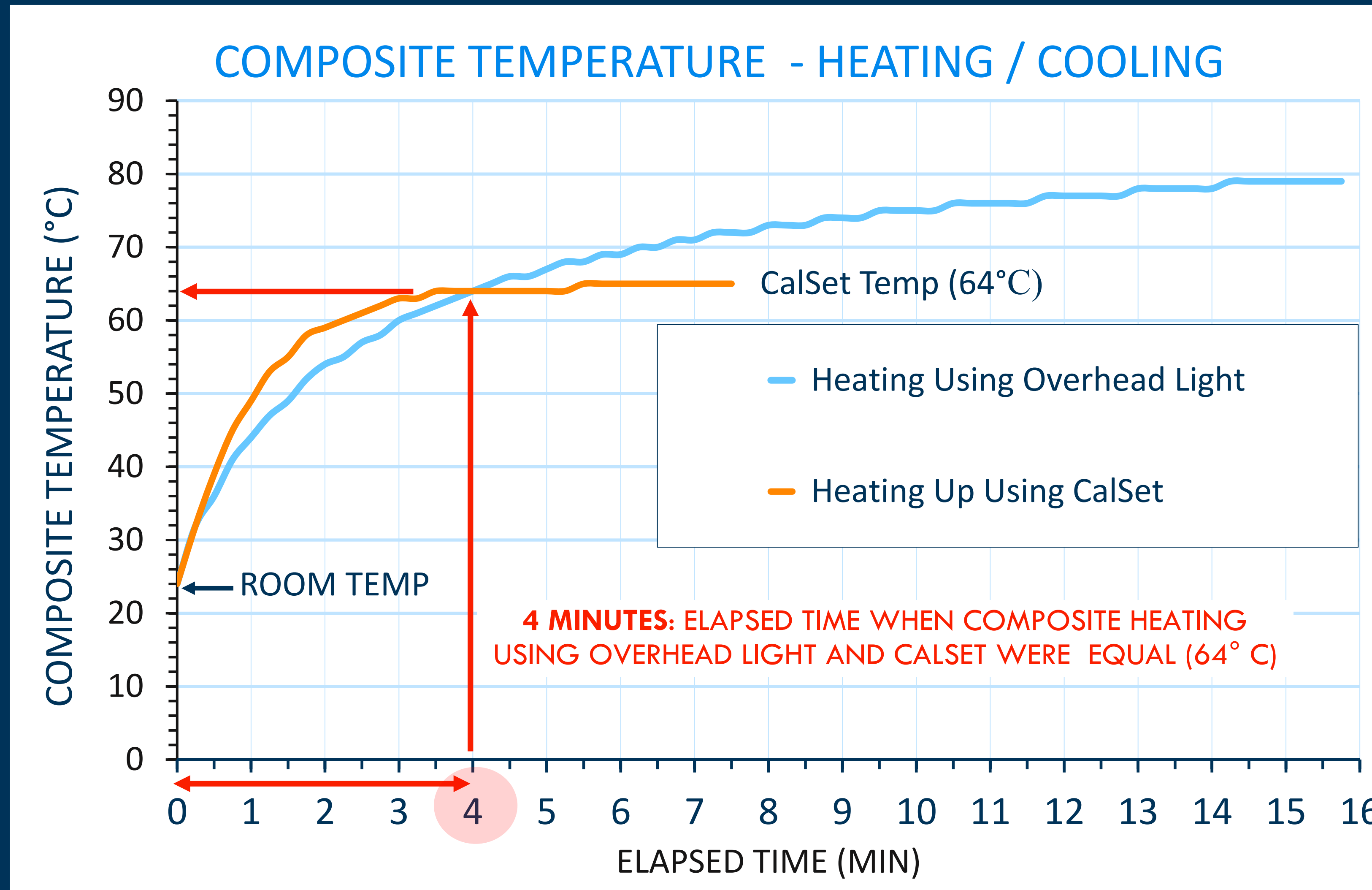
CALSET COMPULE
HEATER

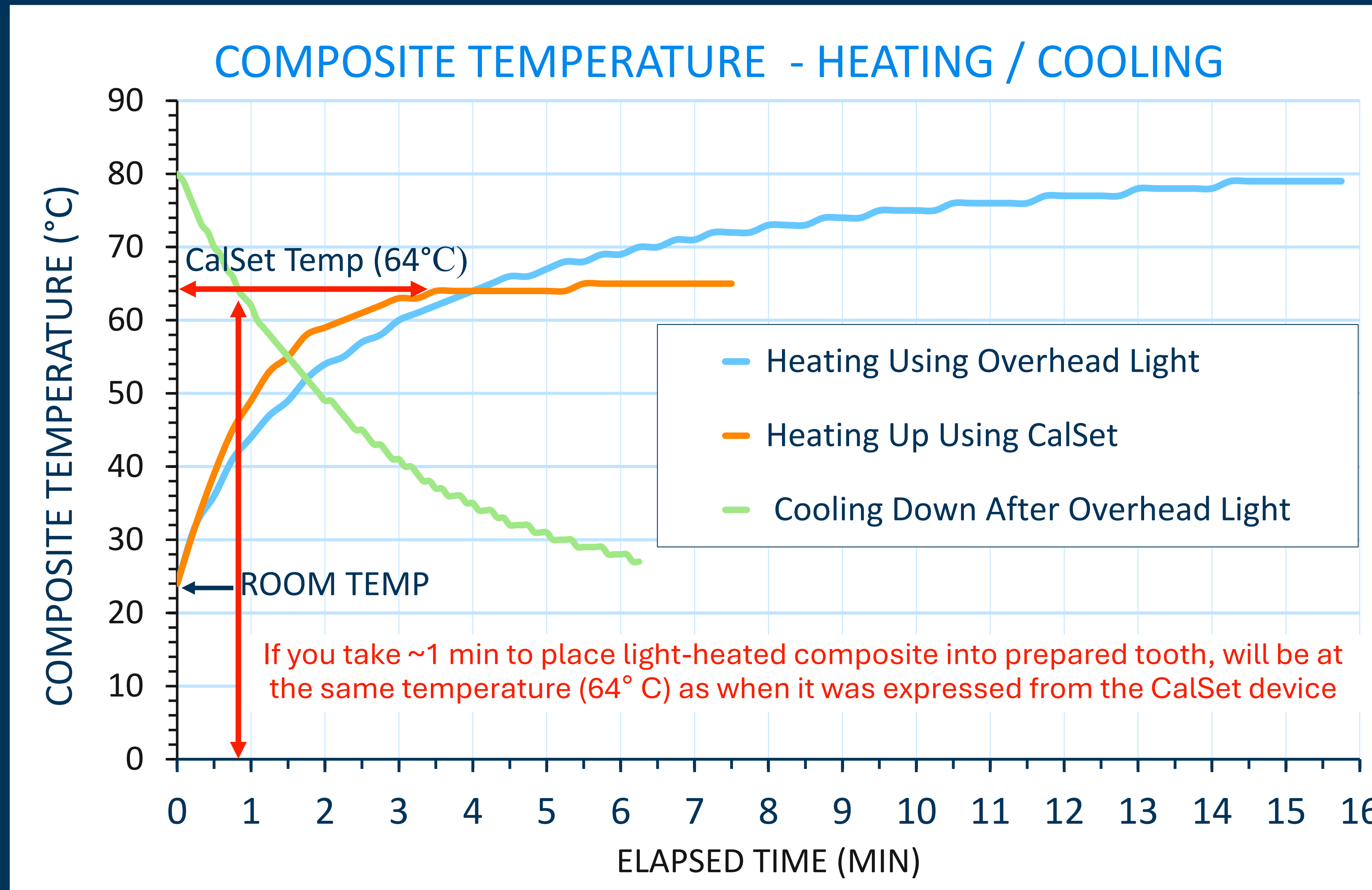


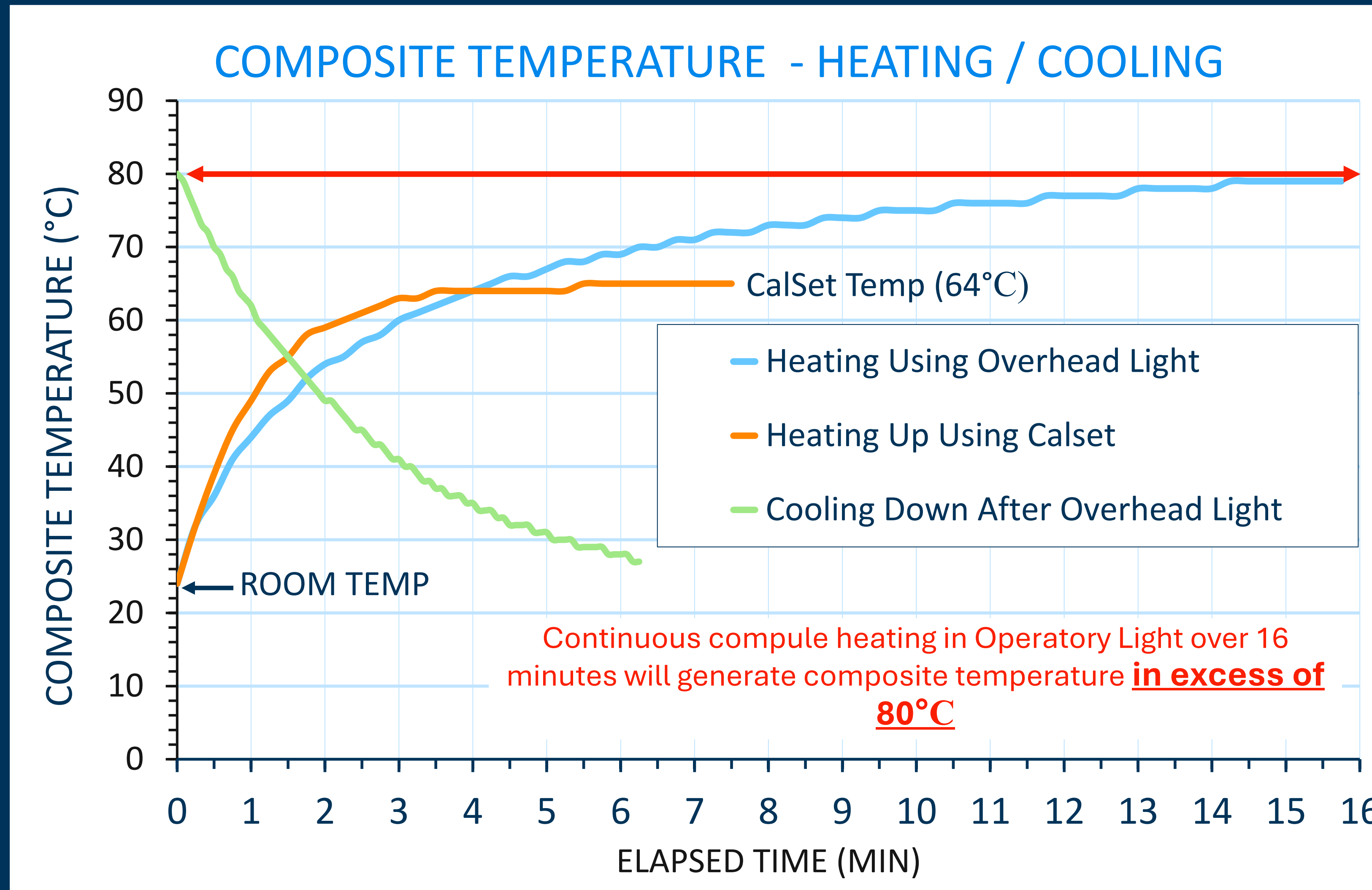
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CALIBRATED DIGITAL
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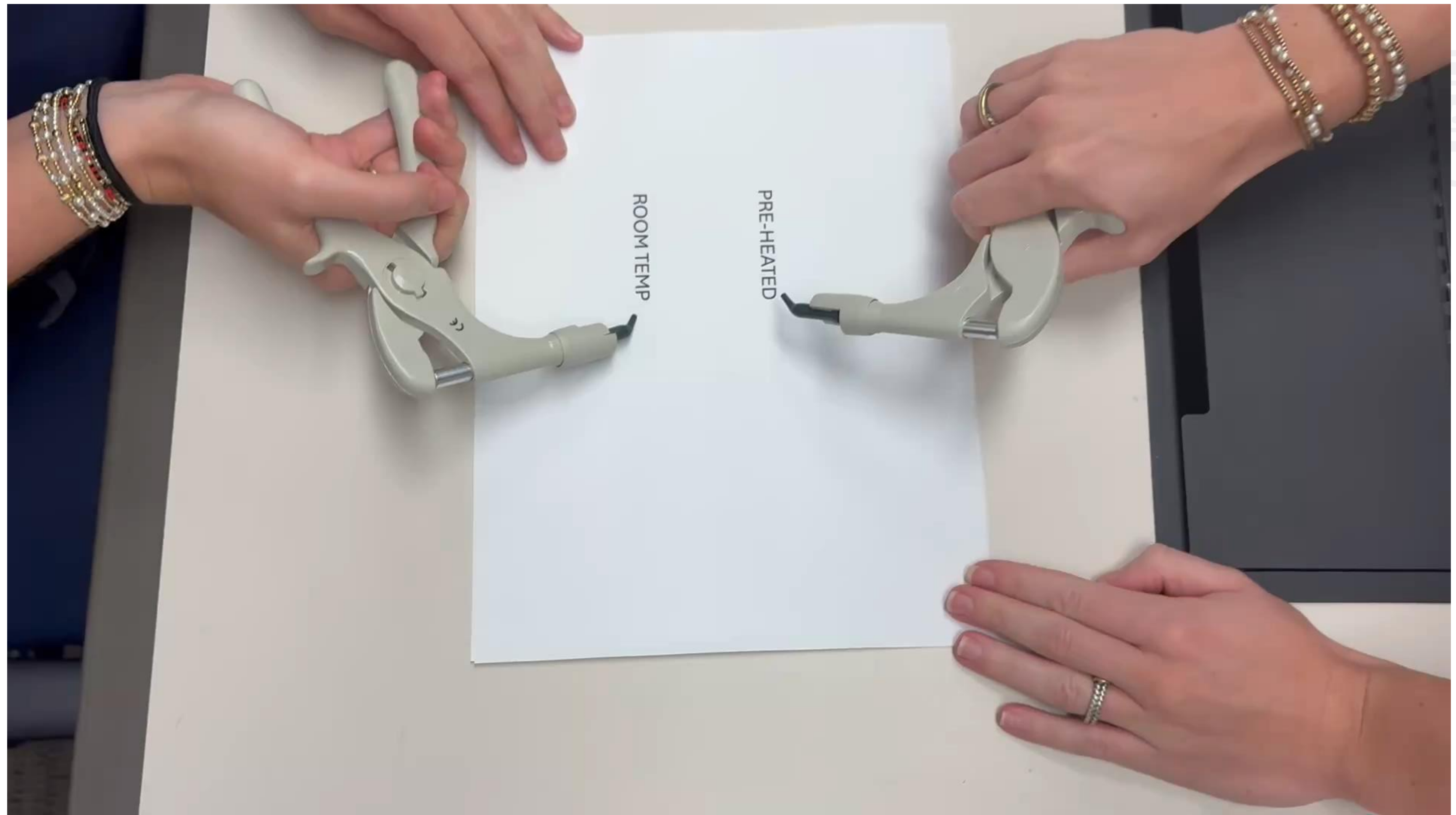


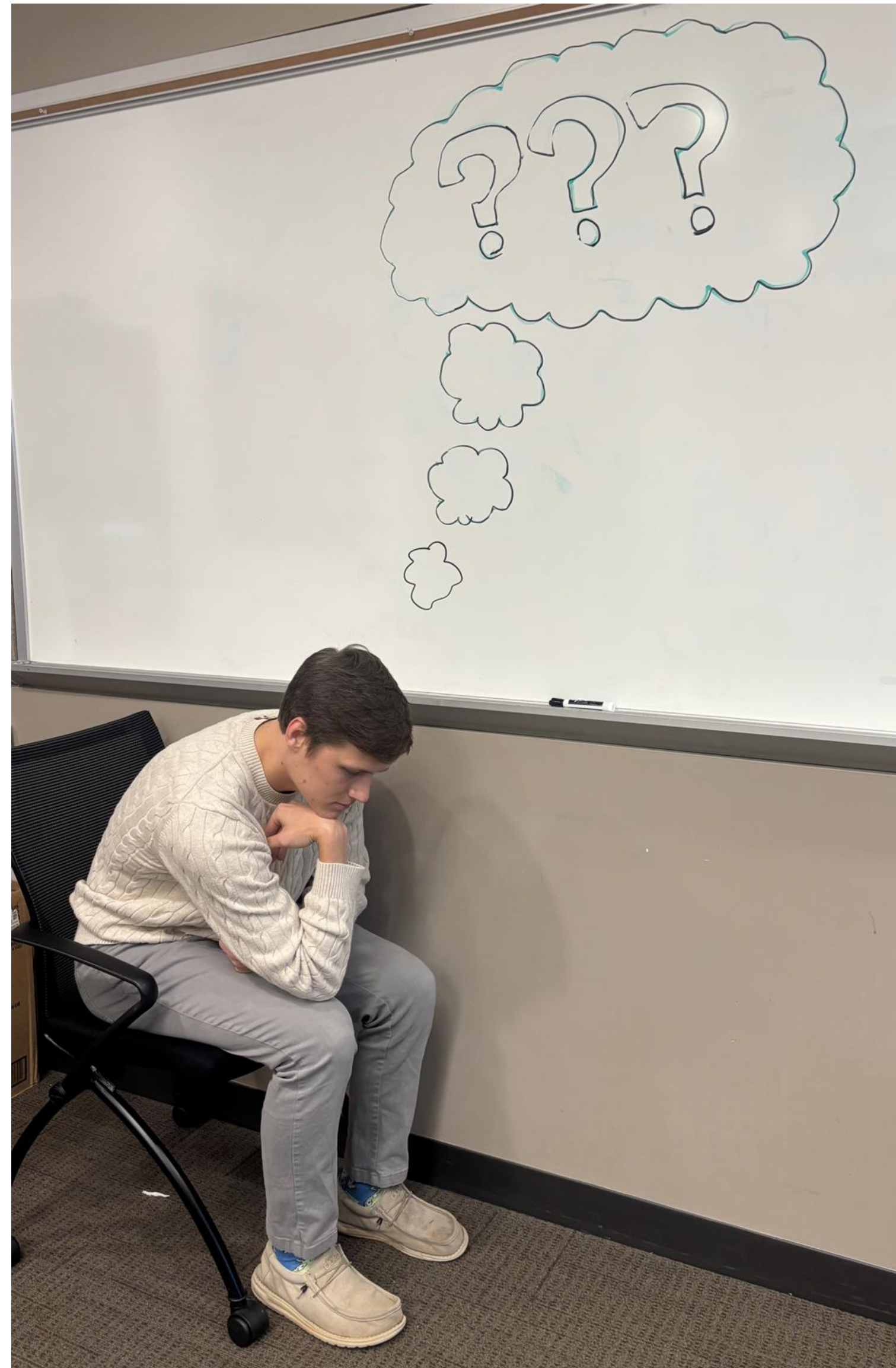














BO1 : State the relationship between composite temperature and viscosity and between resin composite temperature and its extent of monomer conversion when polymerized.

Pre-heating composite lowers the viscosity of packable composite and increases the amount of monomer conversion when polymerized **at that high temperature.**

BO2 : State if heating a conventional resin composite reduces its viscosity to the level of a flowable composite.

Heating conventional resin will decrease the viscosity but not to comparable a level comparable with flowable composite, unless the warmed composite is a special thermo-viscous formulation (VisCalor).





BO3 : State the factors influencing the viscosity and adaptation of preheated composite to the internal preparation walls when it is delivered into clinical preparations.

The factors that influence viscosity and adaption to the internal walls of the prep include increased wettability, decreased monomer elution in bulk fill composites, decreased film thickness and increased stickiness to provide increased adaption and workability.

BO4 : Identify method used to pre-heat resin composite either prior to or subsequent its placement into a clinical preparation and the limitations/advantages to using the various methods

Methods used to preheat resin composite include heated composite dispensers, bench top warmers, digital wax pots and placing them on heat-generating overhead lights. Some limitations to methods are pricing, heating materials in a way that is outside of the manufacturer's recommendations, and increased procedure time.





BO5 : Compare the time required for a resin composite compule contents to reach CalSet-equilvanet temperatures using a halogen operator light determine the halogen light-heated composite's maximum temperature, and state how long it takes for operator light-heated composite to cool to a value similar to that of composite pre-heated using the CalSet device.

It takes approximately four minutes for the halogen light heated composite to reach the 65°C that the CalSet was programmed to, and the maximum temperature of the operator light-heated composite was at least 80°C.

It takes just less than a minute for composite pre-heated using an operator light to cool to a temperature similar to that of composite pre-heated using the CalSet device.

Leaving compule on operator light for undefined time may result in composite being excessively heated

QR CODES FOR LITERATURE REFERENCES (BOX)



BO 1



BO 2



BO 3



BO 4



ALL
REFERENCES



Questions?

